

Power Terminal

Infrastructure: Power Grid

Budget: \$100

Location: Dubai, UAE

Target Year: 2040

Risk Level: LOW

Risk Score: 0.46/100

Climate Resilience Report for Dubai Power Grid

Given the coordinates [25.1867, 55.2515], [25.1855, 55.2515], [25.1855, 55.2509], [25.1866, 55.2510] in Dubai, and considering the recent severe flooding in April 2024 and projected sea level rise, this report outlines site-specific resilience strategies for the power grid infrastructure.

STRATEGY 1

Elevating Power Grid Infrastructure involves designing and constructing power transmission and distribution lines above the projected flood levels. This approach ensures the continuity of power supply during flood events by preventing water from reaching critical components. By utilizing elevated structures, such as poles or towers, the power grid can be protected from inundation, thereby reducing the risk of power outages and associated disruptions.

STRATEGY 2

Implementing Flood-Resistant Power Grid Components entails the use of materials and equipment that can withstand floodwaters, such as waterproof transformers, switchgears, and cables. This strategy focuses on enhancing the resilience of power grid components to minimize damage from flooding, allowing for quicker recovery and reducing the need for extensive repairs. By specifying flood-resistant materials, the power grid can maintain its functionality even when exposed to floodwaters.

STRATEGY 3

Integrating Smart Grid Technologies with Flood Monitoring Systems enables real-time monitoring of flood events and automatic responses to prevent or mitigate damage to the power grid. This strategy leverages advanced technologies, such as IoT sensors and predictive analytics, to detect flood risks and trigger proactive measures, such as load shedding or grid segmentation, to protect the power grid from flooding. By integrating flood monitoring with smart grid technologies, the power grid can respond more effectively to flood events, reducing the risk of power outages and associated disruptions.

RESILIENCE SCORE: 87%

